ic_autolab round 13 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T05:09:59Z at git `cbbedb92c5f124af56dee74c137878aa74629c91`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-342ea5` (`python3 tools/pcc/pcc.py standing PCC-20261104-342ea5`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20 **null**	ksum4f-n8	0.971 [0.931, 1.012]	0.967 [0.925, 1.012]	0.951 [0.895, 1.011]	sel < inc	8.540 / 9.192 / 15.058	8.798 / 9.501 / 15.831	1.583 / 1.698 / 2.733	5.4x / 5.4x / 5.5x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

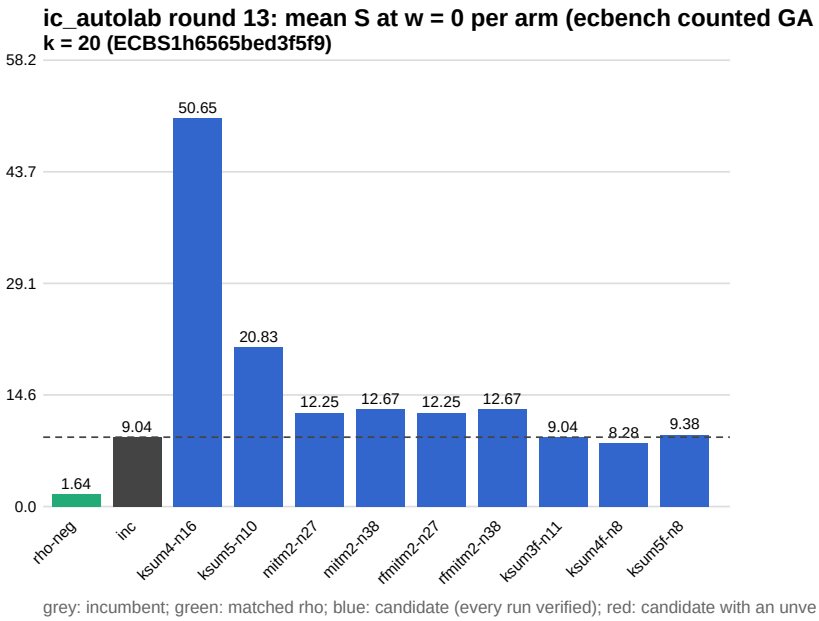
- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.9708 [0.931, 1.012] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h6565bed3f5f9` (run exit 0, verify exit 0, receipt sha256 `6a9fc25aadc2667ddf56f735adc76f6eb557b7cb5f0ef661b094c2a344dde5ea`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.643 / 1.764 / 2.851	matched reference
`inc`	ksum3	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3`	32	32	9.043 / 9.781 / 16.417	incumbent
`ksum4-n16`	ksum4	16	`prime-projected-abscissa:size=16,scan_cap=256`	`ksum:m=4`	32	32	50.645 / 50.903 / 53.219	cross-size transfer neighbourhood: N = 2 x 8 (ksum4 from k = 24)
`ksum5-n10`	ksum5	10	`prime-projected-abscissa:size=10,scan_cap=160`	`ksum:m=5`	32	32	20.827 / 21.047 / 23.031	alternative ksum5 (best S(0) 9.930 at N = 8): N = 1.2 x 8
`mitm2-n27`	mitm2	27	`prime-projected-abscissa:size=27,scan_cap=432`	`mitm:negation_folded=1`	32	32	12.246 / 13.329 / 23.070	alternative mitm2 (best S(0) 11.170 at N = 32): N = 0.85 x 32
`mitm2-n38`	mitm2	38	`prime-projected-abscissa:size=38,scan_cap=608`	`mitm:negation_folded=1`	32	32	12.668 / 13.472	alternative mitm2 (best S(0) 11.170 at

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
						/ 20.705	N = 32): N = 1.2 x 32
`rfmitm2-n27`	rfmitm2	27	`prime-projected-abscissa:size=27,scan_cap=432`	`rootfind-mitm:m=2`	32	32	12.246 / 13.329 / 23.070	alternative rfmitm2 (best S(0) 11.254 at N = 32): N = 0.85 x 32
`rfmitm2-n38`	rfmitm2	38	`prime-projected-abscissa:size=38,scan_cap=608`	`rootfind-mitm:m=2`	32	32	12.668 / 13.472 / 20.705	alternative rfmitm2 (best S(0) 11.254 at N = 32): N = 1.2 x 32
`ksum3f-n11`	ksum3f	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3,free=1`	32	32	9.043 / 9.781 / 16.417	exploration: family ksum3f untried at k = 20, at its parent ksum3's best N = 11
`ksum4f-n8`	ksum4f	8	`prime-projected-abscissa:size=8,scan_cap=128`	`ksum:m=4,free=1`	32	32	8.282 / 8.898 / 14.438	exploration: family ksum4f untried at k = 20, at its parent ksum4's best N = 8
`ksum5f-n8`	ksum5f	8	`prime-projected-abscissa:size=8,scan_cap=128`	`ksum:m=5,free=1`	32	32	9.379 / 9.781 / 13.400	exploration: family ksum5f untried at k = 20, at its parent ksum5's best N = 8



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unve

train screen

Selection

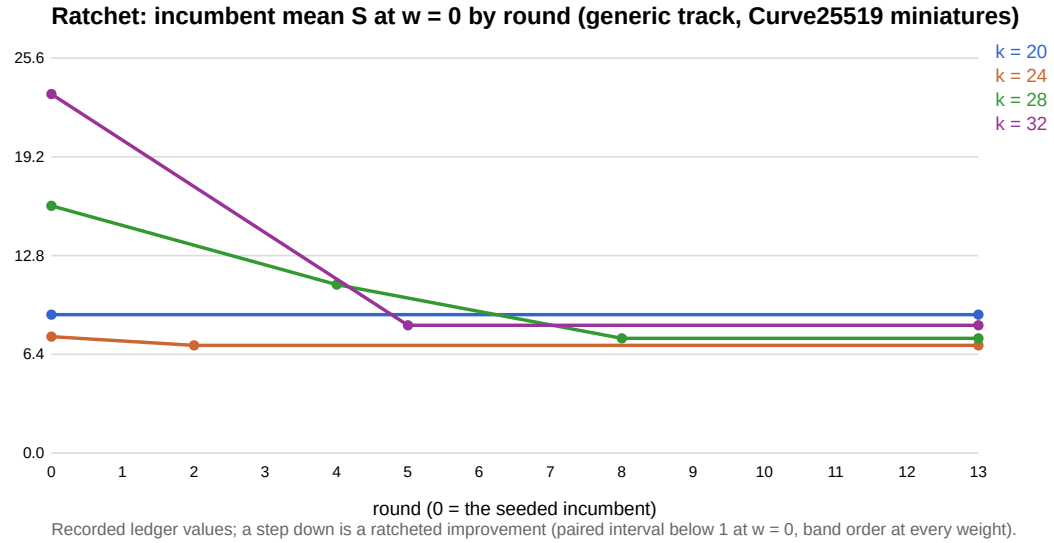
- k = 20: challenger `ksum4f-n8` (Pareto alternative at w = 1: `ksum5f-n8`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	ECBS1ha91af1cc9a88	0	0	a9ef08af15aa14758a3b4ef8732dc03788ecc5efcb2798707bc1c2164ee72b17	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	ksum4-n18 (prime-projected-abscissa:size=18,scan_cap=288 + ksum:m=4)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	ksum4-n26 (prime-projected-abscissa:size=26,scan_cap=416 + ksum:m=4)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `2dedee1fa0e91ee8161c47f825ff60dfaa00e4923cd6b2bb87afdd21bf238b4e`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.